

Auditing Model Substitution in LLM APIs

Why Software Tests Fail and Hardware Attestation Wins

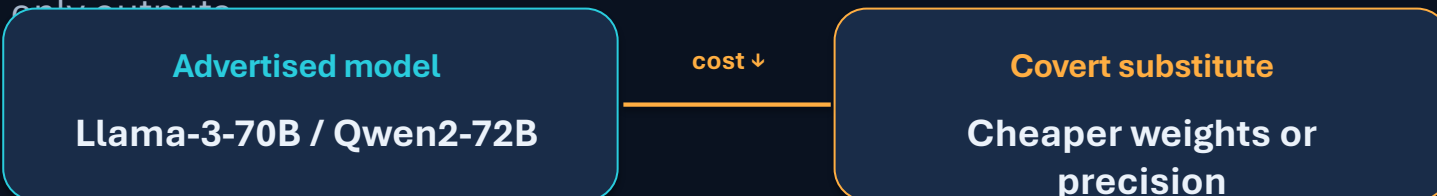
Commercial APIs expose a trust gap: users pay for a named model, but black-box inference provides no proof that the advertised weights executed.

Audience: ML researchers · AI security practitioners · cloud infrastructure engineers

Source: arxiv.org/abs/2504.04715

The Model Substitution Problem: Economic Incentives Drive Covert Swaps

Commercial LLM APIs create asymmetric information:
providers control inference infrastructure, while users see



Formal audit question

$$H_0: y \sim P_{\text{spec}}(y|x)$$

$$H_1: y \sim P_{\text{actual}}(y|x)$$

Can samples alone distinguish them?

1 Smaller model

8B instead of 70B

2 Quantized variant

FP8 / INT8 instead of FP16

3 Fine-tuned update

same family, changed distribution

4 Different family

architecture swap

Implication

The most profitable substitutions are also the hardest to detect: output behavior can remain close while infrastructure cost changes materially.

FP8 Quantization Preserves Most Accuracy — Making Substitution Hard to Detect

Original vs. FP8 scores from the source brief; many FP8 values remain close to original measurements, especially on MMLU and GSM8K.

Model family	Variant	MMLU	GSM8K	MATH	GPQA Diamond
Llama-3-8B-Instruct	Original	62.69 ± 0.18	61.14 ± 3.47	20.65 ± 3.43	22.62 ± 0.22
Llama-3-8B-Instruct	FP8	62.43 ± 0.26	60.90 ± 4.10	14.91 ± 2.49	20.14 ± 0.24
Llama-3-70B-Instruct	Original	78.05 ± 0.08	88.06 ± 1.44	35.69 ± 1.33	29.60 ± 0.51
Llama-3-70B-Instruct	FP8	77.88 ± 0.13	87.35 ± 1.24	35.75 ± 1.16	33.12 ± 0.30
Gemma-2-9b-it	Original	71.86 ± 0.08	81.80 ± 1.35	33.41 ± 0.28	28.64 ± 2.97
Gemma-2-9b-it	FP8	71.92 ± 0.11	79.41 ± 1.14	32.53 ± 0.34	27.81 ± 3.22
Qwen2-72B-Instruct	Original	82.18 ± 0.08	86.72 ± 1.00	37.39 ± 1.41	29.93 ± 2.71
Qwen2-72B-Instruct	FP8	81.98 ± 0.08	86.82 ± 0.97	37.67 ± 1.39	31.08 ± 1.96
Mistral-7B-Instruct-v0.3	Original	59.15 ± 0.10	35.90 ± 4.54	8.94 ± 1.24	21.60 ± 0.17
Mistral-7B-Instruct-v0.3	FP8	58.77 ± 0.13	32.20 ± 4.02	7.68 ± 1.23	22.72 ± 0.19

Detection problem: benchmark-level deltas are small enough that normal evaluation variance and sampling noise can mask substitution.

Higher Temperature Amplifies Performance Gaps but Increases Noise

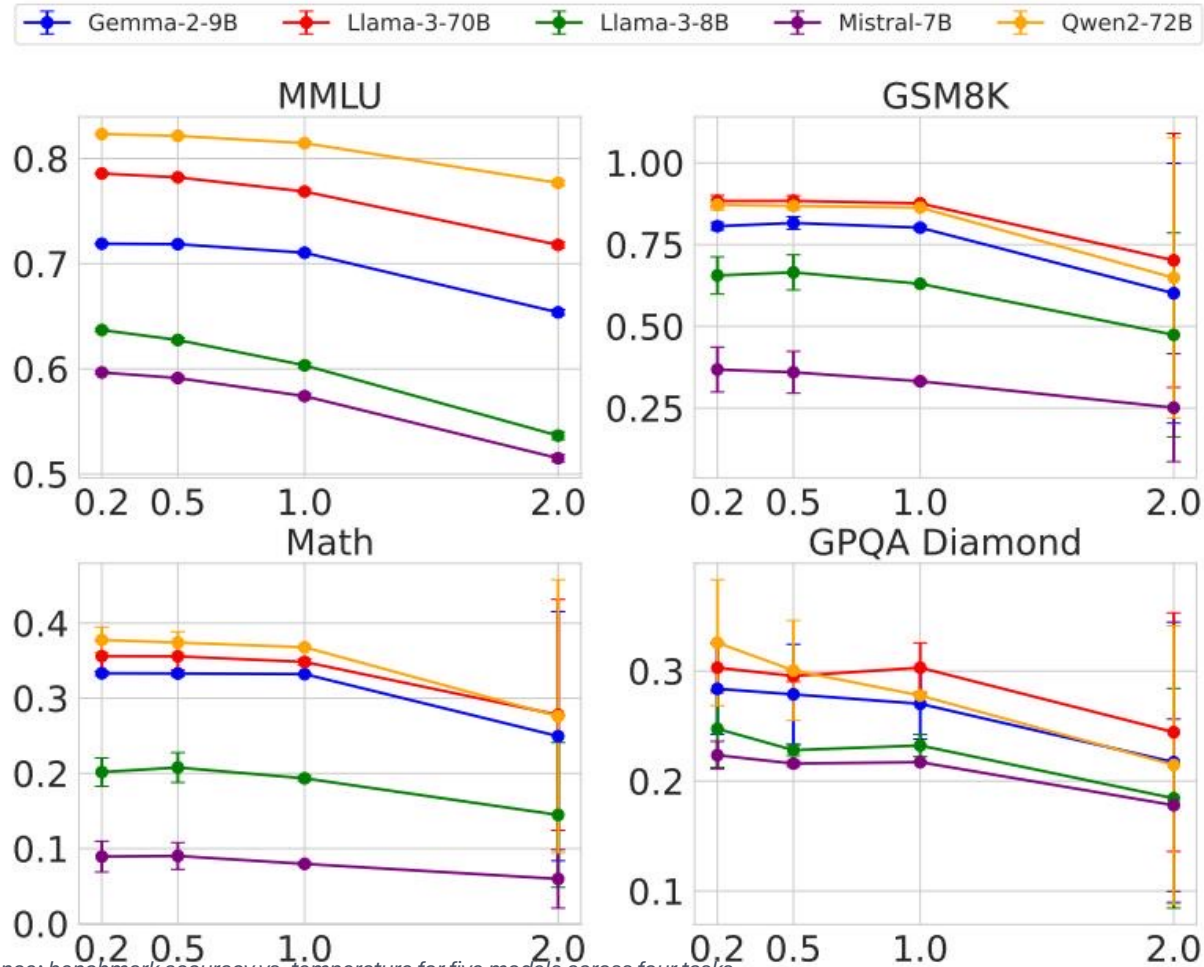


Figure reference: benchmark accuracy vs. temperature for five models across four tasks

Observation

Temperatures 0.2, 0.5, 1.0, and 2.0 show that higher sampling temperature degrades all models' performance.

Audit impact

As temperature rises, the signal from a substituted model is entangled with ordinary generation noise, making statistical separation less reliable.

Practical lesson

Even benchmark-based audits depend on decoding settings; a provider can tune serving behavior to obscure small distributional differences.

Log Probability Fingerprinting Fails Due to Inference Nondeterminism

The same Llama-3-8B model can emit different token log probabilities across production software stacks and GPU hardware.

Model	BERT Acc	T5 Acc	GPT2 Acc	LLM2Vec Acc
Llama3-70B-Instruct-FP8	50.55	50.10	50.45	49.90
Llama3-70B-Instruct-INT8	51.60	49.90	51.30	50.25
Gemma2-9b-it-FP8	49.95	50.30	51.20	49.80
Gemma2-9b-it-INT8	49.00	49.70	51.65	49.55
Mistral-7b-v3-Instruct-FP8	50.55	49.75	48.70	48.75
Mistral-7b-v3-Instruct-INT8	49.70	50.75	50.50	51.15
Qwen2-72B-Instruct-FP8	50.05	50.25	48.75	49.80
Qwen2-72B-Instruct-INT8	50.75	50.55	49.45	50.20

Figure reference: log-probability variation across inference frameworks and hardware for Llama-3-8B

Framework drift

transformers 4.40 vs 4.50

Serving stack drift

vLLM 0.6.2 vs 0.8.2

Hardware drift

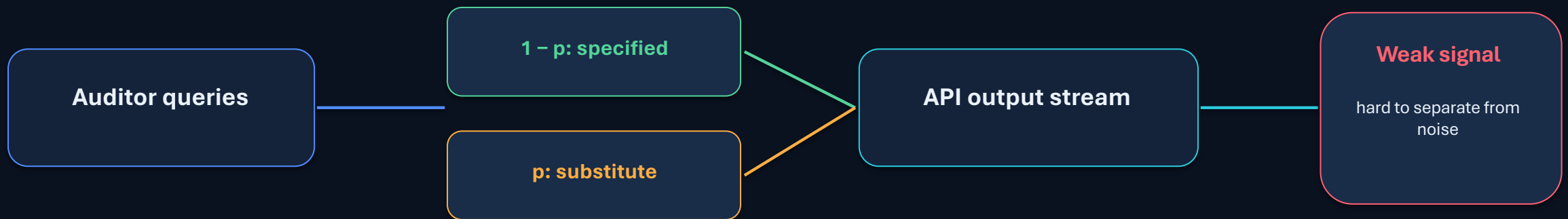
A100 vs H100

Consequence: an auditor cannot reliably tell legitimate infrastructure variation apart from actual model substitution.

Randomized Routing at Low Rates Makes Statistical Detection Query-Prohibitive

A provider can route only a fraction p of requests to a cheaper model, so the observed API distribution approaches the specified model as p decreases.

$$P_{\text{mix}}(y|x) = (1 - p) \cdot P_{\text{spec}}(y|x) + p \cdot P_{\text{sub}}(y|x)$$



As $p \rightarrow 0$

The mixed distribution converges to P_{spec} , reducing observable divergence.

Benchmark evasion

Known benchmark prompts can be routed to the genuine model.

Result

Software-only audits become query-prohibitive in realistic adversarial settings.

Software-Only Auditing Methods Are Fundamentally Unreliable

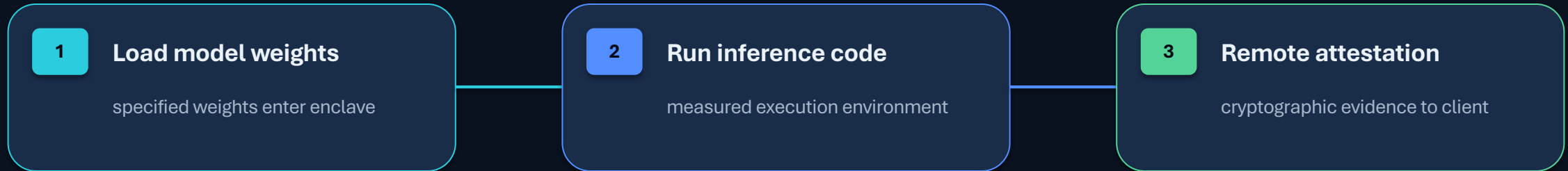
The failure mode is not one weak test; it is a class-wide mismatch between black-box evidence and the integrity claim being audited.

Method type	Weakness	Practical limitation
Text-based statistical tests (e.g., MMD)	Need too many queries; fail against subtle substitutions.	Impractical budgets for continuous production monitoring.
Log-probability methods	Production nondeterminism changes token-level metadata across frameworks and hardware.	False positives: legitimate infrastructure changes can look like substitution.
Adversarial countermeasures	Randomized routing and benchmark evasion reduce or hide the substitution signal.	An adaptive provider can make detection query-prohibitive.

Conclusion: software outputs can support suspicion, but they cannot prove which model weights executed.

Trusted Execution Environments Provide Provable Cryptographic Model Integrity

TEEs shift the audit target from behavior inference to hardware-backed proof of code and data loaded for inference.



Why this wins over software tests

Provable attests code and model data rather than inferring from outputs

Efficient practical today with only modest performance overhead

Robust not defeated by sampling temperature, benchmarks, or randomized routing

Deployment path

- NVIDIA confidential computing on H100 GPUs is cited as a practical option.
- ZKPs are described as computationally prohibitive for large models.
- TEE attestation creates an integrity claim that clients can verify before trusting inference.

Key Takeaways: Close the Verification Gap with Hardware Attestation

Behavioral evidence is not enough; verifiable inference needs cryptographic proof of what executed.

Economic motive

Black-box APIs let providers reduce GPU cost while maintaining advertised pricing.

Quantization camouflage

FP8 barely affects many benchmark scores, making substitution attractive and hard to observe.

Metadata fragility

Log-probability fingerprints break under framework and hardware nondeterminism.

Adaptive evasion

Randomized routing and benchmark evasion undermine software-only audits.

Actionable path

TEEs provide the currently viable route to provable LLM API integrity.

Community call: make attestation a first-class requirement for high-stakes commercial LLM APIs.